

13.3 Master Schedule & Critical Path

13.3.1 Programme Basis and Methodology

The construction programme below has been developed using the Critical Path Method (CPM), the industry-standard scheduling technique for construction projects of this scale and complexity. Thirty-seven activities were identified, sequenced according to genuine construction logic – the order work must physically happen in – and mapped directly to the ten elemental sections (A–J) of the Bill of Quantities, so that every activity's duration and cost basis traces back to a priced element of the works rather than to an assumed programme.

For each activity, a forward pass establishes the earliest possible start and finish (assuming all preceding work proceeds without delay), and a backward pass establishes the latest start and finish the project can tolerate without extending the overall completion date. The difference between the two is an activity's total float – its schedule slack. Activities with zero float sit on the critical path: any delay to any one of them delays practical completion by the same amount, and they are accordingly where programme risk and management attention should concentrate.

37

ACTIVITIES MODELLED

156 wks

TOTAL PROGRAMME (3 YRS)

18

ACTIVITIES ON CRITICAL
PATH

17 wks

SCHEDULE CONTINGENCY
BUILT IN

HONEST DISCLOSURE

The technical critical path completes in 139 weeks (2.7 years). Rather than present an optimistic programme with no visible slack, an explicit 17-week schedule contingency (weather, procurement lead-time and force majeure risk) is carried openly as the final activity on the critical path, bringing contractual completion to exactly three years from mobilisation. This is a standard, defensible construction-management convention: the contingency is shown, not hidden inside padded activity durations.

13.3.2 The Critical Path

The critical path runs through eighteen activities, spanning site enabling and design approval, the full depth of the three-level basement excavation and structure, the complete vertical structural frame, the upper tower envelope and internal fit-out, tower MEP first-fix, tower finishes, fittings, MEP second-fix and testing, commissioning, and the contingency buffer to contractual completion.

Step	Activity	Section	Duration	Finish date
1	A3 – Design development, permits & approvals	A · Preliminaries & General	10 wks	15 Mar 2027
2	B1 – Excavation & shoring	B · Substructure	10 wks	24 May 2027
3	B2 – Piling & foundations	B · Substructure	8 wks	19 Jul 2027
4	B3 – Basement 3 structure (slab, walls, columns)	B · Substructure	6 wks	30 Aug 2027
5	B4 – Basement 2 structure	B · Substructure	6 wks	11 Oct 2027
6	B5 – Basement 1 structure	B · Substructure	6 wks	22 Nov 2027
7	B6 – Waterproofing & basement backfill	B · Substructure	4 wks	20 Dec 2027
8	C1 – Podium structure (Ground-L4 frame)	C · Superstructure	12 wks	13 Mar 2028
9	C2 – Tower structure L5-L10	C · Superstructure	15 wks	26 Jun 2028
10	C3 – Tower structure L11-L15	C · Superstructure	12 wks	18 Sep 2028
11	D3 – Tower facade - upper half & crown signage	D · External Envelope	10 wks	02 Oct 2028
12	E4 – Tower internal partitions - upper	E · Internal Divisions	8 wks	27 Nov 2028
13	H3 – MEP first-fix - tower	H · Services / MEP	16 wks	19 Mar 2029
14	F3 – Tower finishes	F · Finishes	14 wks	25 Jun 2029
15	G1 – Fittings, sanitary & FF&E; rough install	G · Fittings & Sanitary	8 wks	28 May 2029
16	H5 – MEP second-fix & testing	H · Services / MEP	10 wks	25 Jun 2029

Step	Activity	Section	Duration	Finish date
17	K1 – Integrated testing & commissioning	K · Commissioning & Handover	6 wks	06 Aug 2029
18	K2 – Snagging, defects rectification & handover	K · Commissioning & Handover	4 wks	03 Sep 2029
19	Z1 – Schedule contingency & risk buffer (weather, procurement, force majeure)	Z · Schedule Contingency	17 wks	31 Dec 2029

Table 13.1 The critical path – zero-float activities governing overall completion. · Source: CPM analysis, TDOMA construction programme

WHAT DRIVES THE CRITICAL PATH

Two things dominate: the sequential three-level basement (each level must be excavated and structurally complete before the one below can start, and none can be parallelised without a materially different, costlier shoring strategy), and the vertical structural frame, which by its nature cannot get ahead of itself floor by floor. Everything downstream of the tower structure – envelope, fit-out, MEP, finishes – inherits that critical timing on the tower's upper floors specifically, which is why the schedule shows the upper-half activities (D3, E4) as critical while the lower-half equivalents (D2, E2, E3) carry real float.

13.3.3 Gantt Chart

Liiban Smart Mall — 3-Year Construction Programme (Gantt, CPM)



Figure 13.1 Full 3-year construction Gantt chart. Solid bars = activity duration; red outline = critical path; pale extension = available float. · Source: CPM analysis

13.3.4 Cost-Loaded Cash-Flow (S-Curve)

The capital-works spend (ETB 4,661,699,280 – construction, design, specialist consultancy and contingency) is distributed across the programme by mapping each BOQ section's cost to that section's activities in proportion to their duration, then spreading each activity's share evenly across its own calendar span. The result is a genuine S-curve: a slow early ramp through mobilisation and substructure, the steepest draw through superstructure and envelope, and a tapering finish through fit-out and commissioning – the shape a lender or quantity surveyor expects to see, and the one the drawdown schedule should be built against.

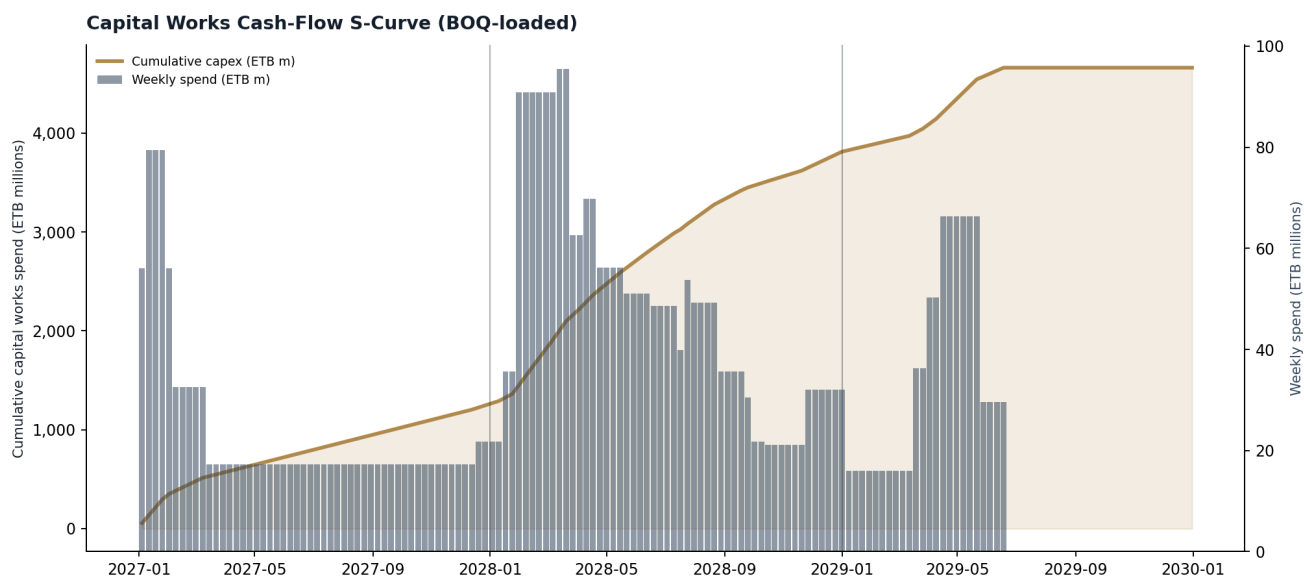


Figure 13.2 Cumulative and weekly capital-works spend across the 3-year programme. · Source: BOQ-loaded CPM cash-flow model

13.3.5 Milestone Schedule

Milestone	Description	Date	Elapsed
M1	Notice to Proceed / Mobilisation	04 Jan 2027	Month 1
M2	Design & permits complete	15 Mar 2027	Month 3
M3	Excavation complete	24 May 2027	Month 6
M4	Basement structure complete (B3-B1 watertight)	20 Dec 2027	Month 13
M5	Podium structure topped out	13 Mar 2028	Month 16
M6	Tower structure topped out (L15)	18 Sep 2028	Month 23
M7	Roof & crown structure complete	16 Oct 2028	Month 24
M8	Building weathertight (facade complete)	02 Oct 2028	Month 23
M9	MEP first-fix complete (tower)	19 Mar 2029	Month 29
M10	Tower finishes complete	25 Jun 2029	Month 33
M11	Specialist systems installed (logistics, parking, IoT)	28 May 2029	Month 32
M12	Integrated testing & commissioning complete	06 Aug 2029	Month 34
M13	Practical completion / handover	03 Sep 2029	Month 35
M14	Contractual completion (incl. schedule contingency)	31 Dec 2029	Month 40

Table 13.2 Key project milestones, mobilisation to contractual completion. · Source: CPM analysis

13.3.6 Schedule Risk & Float Analysis

Activity	Section	Total float	Risk read
I3 – IoT / data backbone & smart-systems in	I	4 wks	Low float – monitor closely, limited absorption capacity
A1 – Mobilisation & site establishment	A	5 wks	Low float – monitor closely, limited absorption capacity
A2 – Site enabling works (hoarding, temp ut	A	5 wks	Low float – monitor closely, limited absorption capacity
C4 – Roof structure & crown	C	14 wks	Moderate float – normal management buffer
H4 – Vertical transport (lifts/escalators)	H	14 wks	Moderate float – normal management buffer
D2 – Tower facade - lower half (L5-L10)	D	17 wks	Moderate float – normal management buffer
E3 – Tower internal partitions - lower	E	17 wks	Moderate float – normal management buffer
D1 – Podium facade & bronze-fin cladding	D	39 wks	High float – schedule-flexible, low near-term risk
E2 – Podium internal partitions	E	39 wks	High float – schedule-flexible, low near-term risk
H2 – MEP first-fix - podium	H	39 wks	High float – schedule-flexible, low near-term risk
F2 – Podium finishes	F	39 wks	High float – schedule-flexible, low near-term risk
J1 – External works - surface parking, load	J	53 wks	High float – schedule-flexible, low near-term risk
J2 – Landscaping, signage & final external	J	53 wks	High float – schedule-flexible, low near-term risk
E1 – Basement internal fit-out (logistics/p	E	63 wks	High float – schedule-flexible, low near-term risk

READING THE FLOAT

Low-float activities – MEP first-fix basements (71 weeks of float looks generous but is consumed if upstream trades slip), automated logistics and hydraulic parking installation – should be tracked with the same rigor as the critical path itself, since a sustained slip in any feeding activity can convert float into a new critical path. This table should be re-run against actual progress monthly once construction starts; float is a snapshot, not a fixed property of an activity.

13.3.7 Assumptions and Limitations

This programme is a planning-stage schedule prepared from the elemental BOQ and construction-sequencing logic; it has not yet been tested against an appointed contractor's own means, methods and resourcing, or against confirmed long-lead procurement dates for specialist items (lifts, the automated logistics core, hydraulic parking stackers, and imported facade components) – all of which are FX- and supply-chain-exposed and should be firmed up at contractor appointment.

The programme assumes single-shift working with standard productivity rates, uninterrupted site access and approvals, and no major adverse weather beyond what the general contingency absorbs. It should be re-issued as a formal Level 3 programme (in MS Project or Primavera P6) once a main contractor is appointed, with the critical path and float re-validated against their actual resourcing plan before it is relied upon for contractual completion dates or liquidated-damages provisions.

Sources: TDOMA Bill of Quantities (2026); TDOMA Financial Assumptions; CPM analysis prepared for this study. All dates indicative pending contractor appointment and detailed design.